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Pre-Generative Artificial Intelligence Prompt Comparison

Abstract

Disclosed herein are system, method, and computer program product embodiments for using generative AI for prompt comparison. The system may receive a prompt. The prompt may include a request for information. The system may identify a stored prompt based on a similarity value between a received prompt for a large language model (LLM) and the stored prompt, the stored prompt including a response generated by the large language model (LLM). The system may generate a response using the LLM if the similarity value between the received and stored prompts is below a predefined threshold. The system may then modify the response by applying a first rule associated with a first designated phrase, and a second rule associated with a second designated phrase, where the first designated phrase comprises a banned phrase and where the second designated phrase comprises a selected phrase.

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Background/Summary

BACKGROUND

[0001] One or more aspects relate to the field of using a large language model (LLM) to analyze prompts and generate responses.

[0002] LLMs are typically used in natural language processing (NLP) to perform a variety of tasks such as text summarization, translation, and speech recognition. LLMs may further be used for text generation. For example, an LLM may generate a response to a prompt or a question. One reason LLMs excel at NLP tasks is that they are trained on vast datasets including billions of tokens (e.g., words, word fragments).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] The accompanying drawings are incorporated herein and form a part of the specification.

[0004] FIG. 1 illustrates an exemplary environment, according to embodiments of the present disclosure.

[0005] FIG. 2 illustrates a prompt engine, according to embodiments of the present disclosure.

[0006] FIG. 3 illustrates a decision tree diagram of an exemplary method for using pre-generative artificial intelligence prompt comparison, according to embodiments of the present disclosure.

[0007] FIG. 4 illustrates a flowchart diagram of an exemplary method for pre-generative artificial intelligence prompt comparison, according to embodiments of the present disclosure.

[0008] FIG. 5 illustrates a flowchart diagram of an exemplary method for identifying similar stored prompts, according to embodiments of the present disclosure.

[0009] FIG. 6 illustrates a flowchart diagram of an exemplary method for generating a new response, according to embodiments of the present disclosure.

[0010] FIG. 7 illustrates a block diagram of an exemplary computer system for implementing one or more aspects of the disclosed embodiments according to various embodiments.

[0011] In the drawings, like reference numbers generally indicate identical or similar elements. Additionally, generally, the left-most digit(s) of a reference number identifies the drawing in which the reference number first appears.

DETAILED DESCRIPTION

[0012] Provided herein are system, apparatus, device, method and/or computer program product embodiments, and/or combinations and sub-combinations thereof, for pre-generative artificial intelligence prompt comparison.

[0013] LLMs are typically used in natural language processing (NLP) to perform a variety of tasks such as text summarization, translation, and speech recognition. LLMs may further be used for text generation. For example, an LLM may generate a response to a prompt or a question. One reason LLMs excel at NLP tasks is that they are trained on vast datasets including billions of tokens (e.g., words, word fragments). However, this means that the LLM may use or combine any of the training tokens when formulating a response. This may lead to inconsistent responses or responses with slight variations. In certain fields, such as finance or wealth management, precise language is required and there is a need to ensure that LLM responses are consistent. Thus, there is a need to leverage LLMs to generate responses while also tailoring them for certain environments or use cases.

[0014] Generative AI (e.g., LLMs) may be useful in communicating data between devices over a network. For example, a bank may host a chat bot that customers may use to request information about their account. The bank may leverage an LLM to interpret and respond to chat bot queries.

As another example, a legal database may allow users to search for legal issues. Here, an LLM may interpret the query and generate a summary of the issue, using relevant statutes and case law. As stated above, LLMs may use any of its training data to formulate a response. This presents a problem in certain environments, where, an LLM is used but precise language is required. A prompt engine, as described below, may be used to receive prompts via a network, leverage an LLM to generate responses, and then filter responses based on one or more rules.

[0015] Certain systems may wish to filter or edit responses before they are sent to client devices over a network. For example, a system may wish to remove certain words or phrases from the response as well as add required language to the response. In some embodiments, systems may desire to use previous responses in order to save computing and network resources. For example, a received prompt may be compared to a set of stored prompts. If a certain degree of similarity between the received prompt and a stored prompt is detected, the system may output a response associated with the stored prompt.

[0016] For example, a first client device may be associated with a database administrator and the client device may send a prompt requesting the number of entries in the database. A system, described in more detail below, may receive the prompt, use an LLM to interpret it and generate a response. This prompt and response may then be stored. A second client device may send a second prompt asking for the same information. In this example, the system may first consult a store of saved prompts, and determine that the second prompt is the same as the first. In this example, the system may respond with the same response, previously generated by the LLM. This operation is beneficial to save computing resources as well as increase consistency of responses.

[0017] As will be discussed further, if none of the stored prompts match the received prompt or lack a required degree of similarity, the LLM may be leveraged to generate a new response to the prompt. LLM may be further leveraged if a predefined amount of time has passed between receiving the current prompt and when the stored prompt was received.

[0018] FIG. 1 illustrates an exemplary environment **100**, according to embodiments of the present disclosure. Environment **100** may include client device **102**, network **104**, prompt engine **106**, and service **108**. Client device **102** may be any entity in communication with prompt engine **106**. Client device **102** may be a computer system such as computer system **700** described with reference to FIG. 7. Client device **102** may be a client system such as a desktop workstation, laptop or notebook computer, netbook, tablet, smart phone, and/or other computing device that may be using an enterprise computing system. Client device **102** may be associated with service **108**. For example, client device **102** may be affiliated with a customer or employee of service **108**.

[0019] Network **104** may be any type of computer or telecommunications network capable of communicating data, for example, a local area network, a wide-area network (e.g., the Internet), or any combination thereof. The network may include wired and/or wireless segments. In some embodiments, network **104** may be a secure network.

[0020] Service **108** may be any resource that client device **102** attempts to access or request data from. For example, service **108** may be a financial institution (e.g., a bank) and client device **102** may belong to a customer of the financial institution. Although a single service **108** is depicted, environment **100** may include multiple services **108**.

[0021] Prompt engine **106** may be implemented using one or more servers and/or databases. In some embodiments, prompt engine **106** may be implemented using a computing device such as a desktop workstation, laptop or notebook computer, netbook, tablet, smart phone, and/or other computing device. In some embodiments, prompt engine **106** may be implemented as an application in an enterprise computing system and/or a cloud-computing system. In some embodiments, prompt engine **106** may be a computer system such as computer system **700** described with reference to FIG. 7. Although a single prompt engine **106** is depicted, environment **100** may include multiple prompt engines **106**. For example, each service **108** may be connected to a corresponding prompt engine **106**. In some embodiments, a single prompt engine **106** may be

used to respond to prompts associated with multiple services **108**.

[0022] FIG. 2 illustrates prompt engine **106**, according to embodiments of the present disclosure. Prompt engine **106** may include communication device **210**, large language model (LLM) **212**, data store **214**, and filer device **216**. Communication device **210** may be configured to communicate with client device **102** and service **108**, via network **104**. Communication device **210** may comprise any suitable network interface capable of transmitting and receiving data, such as, for example a modem, an Ethernet card, a communications port, or the like. Communication device **210** may be able to transmit data using any wireless transmission standard such as, for example, Wi-Fi, Bluetooth, cellular, or any other suitable wireless transmission.

[0023] Prompt engine **106** may be configured to receive prompts and generate responses to communications over network **104**. In some embodiments, prompts may be received at prompt engine **106** from client device **102**. When a prompt is received, prompt engine **106** may first query data store **214** to determine if data store **214** includes a stored prompt that is similar to the received prompt.

[0024] Data store **214** may be a memory device used to store data. Data store **214** may include various pieces of information such as previously received prompts and corresponding generated responses. Data store **214** may be categorized according to any method or schema. In some embodiments, data store **214** may be organized by service **108**. Here, data store **214** may group prompts and their respective responses under the service **108** associated with the prompt. For example, a first service **108** may be associated with a bank, and a second service **108** may be associated with an online retailer. Since the prompts and responses to each of these services **108** may be different, it may be beneficial to keep the prompts and responses separate. This will help ensure that prompt engine **106** will return the most accurate response. As will be discussed further below, responses may be generated by LLM **212**.

[0025] Prompt engine **106** may compare the received prompt to prompts stored at data store **214**. Prompt engine **106** may use any means to make the comparison. In some embodiments, prompt engine **106** may convert the received prompt and the stored prompts to numerical vectors to perform the comparison. Once converted, prompt engine **106** may apply any vector similarity algorithm such as cosine similarity, Euclidean similarity, or dot product similarity. Prompt engine **106** may select the stored prompt with the highest similarity to the received prompt, and use the response corresponding to the stored prompt.

[0026] As stated above, prompt engine **106** may select a stored prompt with the highest similarity to the received prompt. In some embodiments, prompt engine **106** may set a similarity threshold. For example, prompt engine **106** may require that the stored prompt be at least 90% similar to the received prompt. If none of the stored prompts at data store **214** are above the threshold, prompt engine **106** may use LLM **212** to generate a new response.

[0027] LLM **212** may be a machine learning model represented by a set of weights. The weights may represent features that LLM **212** is designed to learn. LLM **212** may use any suitable machine learning architecture. For example, LLM **212** may be a transformer model. In some embodiments, prompt engine **106** may include a different LLM **212** for each service **108**. In some embodiments, prompt engine **106** may include a single LLM **212** used for each service **108**. In some embodiments, LLM **212** may be a single model, with different sets of weights for each service **108**. LLM **212** may be trained to interpret prompts and generate responses based on the prompt. For example, the prompt may be a question and the response may be an answer. The prompt and response may include text, audio, video, or a combination thereof. For example, a prompt may be a written question such as, “How much money is in my bank account?” and LLM **212** may generate a response such as “There is \$100 in your bank account.”

[0028] LLM **212** may be configured to customize the response based on client device **102**. Data store **214** may include information on client device **102** and its associated user such as: name, address, occupation, education history, device information, device identifier, and previously

received prompts and responses. Data store **214** may also include prompts previously received from client device **102**. In some embodiments, data store **214** may further include client device's **102** interactions with service **108**. LLM **212** may use this information to tailor the generated response. [0029] For example, LLM **212** may be trained to include a greeting with the name of the user associated with client device **102** in each response. In this example, LLM **212** may access data store **214** to retrieve the name of the user associated with client device **102** to include within the response. As another example, client device **102** may send a prompt asking information regarding their bank account. LLM **212** may be configured to generate a template response such as “There is [BLANK] in your bank account.” Prompt engine **106** may retrieve information specific to client device's **102** bank account. If the information is at data store **214**, prompt engine **106** may use the information from data store **214** to fill in the template. In some embodiments, the data may not be at data store **214**. Here, prompt engine **106** may communicate with service **108** to obtain the information and fill in the template.

[0030] LLM **212** may use data at data store **214** for training. For example, LLM **212** may periodically train on data at data store **214** in order to increase its performance. Training data at data store **214** may be organized into examples. Each example may include a received prompt, the response sent to client device **102** (e.g., the actual response), and a label. The label may be a binary value such as 0 if the response was incorrect, and 1 if it was correct. The label may be a continuous value such as 10% correct, 50% correct, or 90% correct. In some embodiments, the label may be received from client device **102**. In some embodiments, the label may be received from service **108**.

[0031] LLM **212** may train by inputting the prompt and generating a response. The label and the actual response may be used to evaluate the generated response and update LLM **212**. For example, if the label indicates the actual response successfully addressed the prompt, LLM may be optimized to generate a response similar to the actual response. If the label indicates the actual response failed to address the prompt, LLM **212** may be trained to generate a different response.

[0032] As will be discussed further below, responses may be edited to remove and include certain language. LLM **212** may therefore be trained to produce responses that require little to no alteration. This may be accomplished by applying a set of rules corresponding to banned and required words or phrases to the generated response. LLM **212** may be updated based on whether the generated response included banned words or phrases, and failed to include required words or phrases.

[0033] In some embodiments, information at data store **214** may be in its original format. For example, text-based prompts and responses may be stored as text, whereas image-based prompts and responses may be stored as images. In some embodiments, data store **214** may maintain the information as numerical vectors. Maintaining information at data store **214** as numerical vectors is beneficial to increase processing speed. As stated above, data store **214** may be searched to identify similar prompts and return the respective response. The search may be based on similarity between the received prompt and the stored prompts at data store **214**.

[0034] Prior to outputting a response, prompt engine **106** may send the response to filter device **216**. Filter device **216** may apply a set of rules to the response. Rules may include banned words, banned phrases, or a combination thereof, that may be removed from the response. The rules may also include selected or designated phrases that may be added to the response.

[0035] For example, filter device **216** may be configured to remove words from the response that are on a list of banned words. Filter device **216** may be further configured to remove phrases on a list of banned phrases. Filter device **216** may be further configured to add required words and/or phrases to a response. For example, a response to a prompt may require certain disclosure information. In this example, filter device **216** may be configured to add the disclosure information to the response. Prompt engine **106** may determine which rules are applied to the response.

[0036] For example, certain rules may be associated with certain services **108**. Therefore, service

108 specific rules may be applied when the prompt and/or response involves the certain service **108**. For example, client device **102** may be a banking customer and service **108** may be the bank. Client device **102** may submit a prompt requesting information regarding their bank account. Prompt engine **106** may generate a response. However, prior to outputting the response, prompt engine **106** may use filter device **216** to consult a set of rules defined by the bank (e.g., service **108**). The bank may have defined a rule requiring certain disclosure materials be included in each response. Therefore, filter device **216** may add the disclosure provided in the rule, to the response. These rules may be unique to service **108**. An additional request associated with a different service **108** may use a different set of rules. Allowing service **108** to define its own set of rules further ensures that responses from prompt engine **106** remain consistent. Additionally, prompt engine **106** may include rules that apply to all responses. For example, prompt engine **106** may include a rule preventing certain profanity from being included in responses.

[0037] Additionally, filter device **216** may apply rules that remove personally identifiable information (PII) from responses. PII may be a social security number (SSN), bank account number, passport number, driver's license number, credit card number, debit card number, date of birth, name, address, or phone number. Prompt engine **106** may configure filter device **216** to search for additional PII examples.

[0038] Each type of PII may have a rule determining whether it is removed from the response. For example, a name may be allowed to be part of a response (e.g., in a greeting) whereas an SSN is always removed. Similar to the rules previously discussed, each service **108** may define its own rules regarding PII. Filter device **216** may use one or more regular expressions to detect the PII and remove it from a response. In some embodiments, filter device **216** may leverage LLM **212** to detect PII. LLM **212** may be used to label each word and/or phrase of a response. The label may correspond to whether the word and/or phrase constitutes PII. Filter device **216** may remove portions of the response that are labeled PII.

[0039] In some embodiments, LLM **212** may train using the rules implemented by filter device **216**. For example, data store **214** may include a training data set of rules currently enforced along with examples of each rule. Each example may include a response that is either in conformity with the rule or in violation of the rule. For example, a first rule may include a list of banned words. The first rule may have two accompanying examples. The first example may include one or more of the banned words along with a label. The second example may not include any banned words defined in the rule. LLM **212** may be trained to identify whether any of the banned words, defined in the first rule, are present in each example. Each example may have a corresponding label. The label may be defined with various degrees of precision. In some embodiments, the label may be a binary value denoting whether the example response violates the rule. In some embodiments, the label may include each word or phrase in violation of the rule, and its location within the response. Training LLM **212** to learn rules implemented by filter device **119** will increase prompt engine's **106** performance and efficiency in analyzing and responding to prompts.

[0040] In some embodiments, prompt engine **106** may save the received prompt and the output of filter device **216** to data store **214**. This may be beneficial so if a future prompt is similar to the received prompt, the same response may be used. Additionally, this information may be used to train LLM **212**. The output of filter device **216** may be sent to client device **102** via network **104**.

[0041] FIG. 3 illustrates a decision tree diagram of an exemplary method **300** for using pre-generative artificial intelligence prompt comparison, according to embodiments of the present disclosure. As shown in FIG. 3, method **300** begins at step **310** by prompt engine **106** receiving a prompt. The prompt may be received by client device **102** via network **104**. The prompt may include text, audio, images, video, or any combination thereof. An example prompt may be, "How much money is in my bank account?"

[0042] At step **320**, prompt engine **106** determines whether a similar prompt may be located at a data store. The data store may be data store **214**. Prompt engine **106** may identify a similar prompt

by computing the similarity between the received prompt and each stored prompt at data store **214**. In some embodiments, prompt engine **106** may convert the received and stored prompts to numerical vectors. Prompt engine **106** may then calculate a vector similarity between the received prompt and stored prompt vectors. The vector similarity may be used as a similarity measure of the prompts. Prompt engine **106** may determine a threshold similarity value. For example, prompt engine **106** may require 80% similarity between the received and stored prompts. If a similar prompt is identified, method **300** continues to step **330**, otherwise method **300** continues to step **340**.

[0043] At step **330**, prompt engine **106** retrieves the response associated with the stored prompt. The response may have been generated by LLM **212**. In some embodiments, prompt engine **106** may edit the retrieved response. Prompt engine **106** may insert or replace words or phrases so that the response is tailored to the requesting entity (e.g., client device **102**). For example, the response may include an identifier and prompt engine **106** may include an identifier corresponding to client device **102**. Retrieving a response previously generated is beneficial because it results in consistency across responses.

[0044] At step **340**, prompt engine **106** applies an LLM to the prompt to generate a response. The LLM may be LLM **212**. As stated above, the LLM may be trained to perform natural language tasks such as language generation. Here, the LLM may input the prompt, analyze it, and output a response. In some embodiments, the LLM may output multiple responses according to a probability distribution. For example, each output may have a probability score corresponding to the LLM's confidence that the output is the correct one. Prompt engine **106** may use the response with the highest probability score.

[0045] At step **350**, prompt engine **106** determines whether any banned words or phrases are in the response. Prompt engine **106** may use filter device **216** to perform the determination. Filter device **216** may consult a list of banned words or phrases. The list may be stored at filter device **216** or data store **214**. Prompt engine **106** may define any list of banned words or phrases. The list of banned words and phrases may be organized into rules. For example, a first rule may include a first set of banned words and phrases, and a second rule may include a second set of banned words and phrases. As an example, a rule may ban profanity, and thus include a list of profane words and phrases to remove. A separate rule may be concerned with leaking PII. Therefore, the rule may include PII that should be removed from the response. In some embodiments, different rules may be defined by and associated with different services **108**. If a banned word or phrase is detected, method **300** continues to step **360**, otherwise method **300** continues to step **370**.

[0046] At step **360**, prompt engine **106** applies a response filter to remove the identified banned words and/or phrases. Prompt engine **106** may use filter device **216** to remove the banned words and/or phrases.

[0047] At step **370**, prompt engine **106** determines whether a required word and/or a required phrase is missing from the response. Prompt engine **106** may consult a list of required words or phrases. The list may be stored at data store **214**. The required words or phrase may also be organized as rules. In some embodiments, the rules may be associated with services **108**. For example, certain services **108** may require certain words or phrases to be included within responses. Some rules may be associated with all responses, regardless of the service **108**. For example, prompt engine **106** may define a rule that all responses should include a date/time field. If a required word or phrase is missing, method **300** continues to step **380**, otherwise method **300** continues to step **390**.

[0048] At step **380**, prompt engine **106** applies a response filter to add the required words and/or phrases to the response. Prompt engine **106** may use filter device **216** to add the words and/or phrases.

[0049] At step **390**, prompt engine **106** transmits the response. The response may be transmitted to client device **102** via network **104**. In some embodiments, prompt engine **106** may further save the

received prompt and generated response at data store **214**. This may be useful to generate future responses as well as to train LLM **212**.

[0050] FIG. **4** illustrates a flowchart diagram of an exemplary method **400** for pre-generative artificial intelligence prompt comparison, according to embodiments of the present disclosure. Method **400** may be used to receive a prompt and leverage an LLM to respond to the prompt. As shown in FIG. **4**, method **400** begins at step **410** by prompt engine **106** receiving a prompt for a large language model (LLM) from a client device. The LLM may be LLM **212**. The client device may be client device **102**.

[0051] At step **420**, prompt engine **106** identifies a stored prompt based on a similarity value between the received prompt and the stored prompt. The stored prompt may have an associated stored response generated by the LLM. The similarity value may be determined by converting the received prompt and stored prompt into vector representations and then calculating the vector similarity.

[0052] At step **430**, prompt engine **106** modifies the response by applying a first rule associated with a first designated phrase, where the first designated phrase comprises a banned phrase. In some embodiments, the banned phrase may be a single word or a collection of words (e.g., a phrase). In some embodiments, prompt engine **106** may use filter device **216** to modify the response. For example, filter device **216** may remove any banned words and/or phrases detected in the response.

[0053] At step **440**, prompt engine **106** modifies the response by applying a second rule associated with a second designated phrase, where the second designated phrase comprises a selected phrase. The selected phrase may also be a single word or a collection of words (e.g., a phrase). The selected phrase may be required to be part of the response. In some embodiments, prompt engine **106** may use filter device **216** to modify the response. For example, filter device **216** may add any required words and/or phrases to the response.

[0054] At step **450**, prompt engine **106** modifies the response based on a previous prompt and previous response from the client device. The client device may be client device **102**. The previous prompt may be located at data store **214**. The previous response may be associated with the previous prompt at data store **214**. Prompt engine **106** may identify the previous prompt by searching data store **214** based on an identifier associated with client device **102**. Prompt engine **106** may modify the response by including information related to client device **102** and/or a user associated with client device **102**.

[0055] For example, the prompt received at step **410** may only include a numeric identifier associated with client device **102**. However, the previous response may include a name of a user associated with client device **102**. Here, prompt engine **106** may modify the response by inserting the name of the user associated with client device **102**.

[0056] In some embodiments, prompt engine **106** may not modify the response if a predefined amount of time has passed between the received prompt and previous prompt. For example, if a year has passed between the received prompt and the previous prompt, prompt engine **106** may not reference the previous prompt since data associated with it may have changed.

[0057] In some embodiments, prompt engine **106** may not modify the response if it detects the modification no longer applies. For example, client device **102** may have belonged to a first employee when the previous prompt was received. Subsequently, client device **102** may have been transferred to a second employee who may have sent the current prompt. Prompt engine **106** may detect that the user associated with client device **102** has changed, and thus may not reference the previous prompt to modify the current prompt.

[0058] At step **460**, prompt engine **106** transmits the modified response to the client device. As stated above, the client device may be client device **102**. Prompt engine **106** may use communication device **210** to transmit the response via network **104**. In some embodiments, prompt engine **106** may additionally save the prompt and modified response to data store **214**. This

is beneficial so that the prompt and modified response may be referenced in responding to future prompts.

[0059] FIG. 5 illustrates a flowchart diagram of an exemplary method **500** for identifying similar stored prompts, according to embodiments of the present disclosure. Method **500** may be used by prompt engine **106** to identify previously received, stored prompts. The responses associated with the previously received prompt may then be used to respond to the current prompt. This process is beneficial to ensure that responses to similar or related prompts are consistent. Method **500** may include additional details related to **420** as described with reference to method **400**.

[0060] As shown in FIG. 5, method **500** begins at step **510** by prompt engine **106** converting the received prompt to a vector representation. In the vector representation, each word or part of a word may be represented by a number. In some embodiments, prompt engine **106** may use a lookup table to convert each word to a number. In some embodiments, prompt engine **106** may use a tokenization algorithm to generate the vector representation. The result of the tokenization algorithm may be that similar words have similar vector representations. This may be beneficial so that the meaning of the words is captured via their vector representations. For example, words that are synonyms (e.g., ocean and lake) should have closer vector representations than words that are unrelated (e.g., ocean and computer).

[0061] At step **520**, prompt engine **106** converts the stored prompt to a vector representation. Prompt engine **106** may use the same technique used in **510** above.

[0062] At step **530**, prompt engine **106** calculates the similarity value between the vector representation of the received prompt and the vector representation of the stored prompt. The similarity value may be determined by any vector similarity algorithm. Prompt engine **106** may calculate the similarity value using cosine similarity, Euclidean similarity, or dot product similarity.

[0063] FIG. 6 illustrates a flowchart diagram of an exemplary method **600** for generating a new response, according to embodiments of the present disclosure. Prompt engine **106** may use method **600** in order to update stored responses and subsequently retrain an LLM using the updated stored responses. As stated above, data communicated in environment **100** may rapidly change. Changes may apply to the data itself, its format, included fields, as well as elements that are required to be communicated and elements that are barred from being communicated. For example, service **108** may be a financial institution. In the financial industry, required disclosures may change resulting from new laws or regulations. Therefore, it's desirable to generate new responses after a certain amount of time has passed in order to capture these updates. In this example, method **600** may be used to detect a predefined amount of time has passed between receiving the current prompt and when the stored prompt was received, and therefore that a new response should be generated.

[0064] During the time window between when the stored response was created and the latest prompt received, LLM **212** may have been updated according to any new laws, rules, and/or regulations. Therefore, creating a new response would also conform to those updates. For example, if the identified stored prompt was received six months prior to the current prompt, prompt engine **106** may be configured to leverage LLM **212** to create a new response. Method **600** may include additional details related to **420** as described with reference to method **400**.

[0065] As shown in FIG. 6, method **600** begins at step **610** by prompt engine **106** determining that a time period between the received prompt and the stored prompt is greater than a predefined time threshold. The time period may be determined by calculating the difference between timestamps of the received prompt and stored prompt. The predefined time threshold may be any time value such as one week, one month, or one year. In some embodiments, service **108** associated with the prompt may define the predefined time threshold and communicate it to prompt engine **106** via network **104**.

[0066] At step **620**, prompt engine **106**, in response to the determination, generates a new response by applying the LLM to the received prompt. The LLM may be LLM **212**. As stated above, the

LLM may be trained to perform natural language tasks such as language generation. Here, the LLM may input the prompt, analyze it, and output a response. In some embodiments, the LLM may output multiple responses according to a probability distribution. For example, each output may have a corresponding probability score corresponding to the LLM's confidence that the output is the correct one. Prompt engine **106** may select the response with the highest probability score. [0067] As described above with regard to steps **430**, **440**, and **450**, prompt engine **106** may then use filter device **216** to modify the response. For example, filter device **216** may use a set of rules to remove banned words and/or phrases, and add required words and/or phrases. Filter device **216** may further modify the response based on information associated with the client device (e.g., client device **102**).

[0068] At step **630**, prompt engine **106** receives a label corresponding to the response. The label may indicate how effective the response was at addressing the prompt. The label may be received from client device **102** or service **108**. The label may be a binary, discrete, or continuous value. In some embodiments, the label may include a ground truth response. For example, LLM **212** may have generated an incorrect response to the prompt. In response, client device **102** or service **108** may send prompt engine **106** a ground truth response (e.g., a correct response) so that LLM **212** may be retrained. As an illustration, a prompt such as "What is the capital of the United States?" may be received. LLM **212** may generate an incorrect response, "Baltimore." Client device **102** and/or service **108** may respond with a label including "Washington, D.C." Providing the ground truth or correct response to a given prompt will allow LLM **212** to be retrained and provide more accurate responses in the future.

[0069] At step **640**, prompt engine **106** adds the received prompt, the modified response, and the label to a training data store. In some embodiments, the training data store may be part of data store **214**. In some embodiments, entries at the training data store may be overwritten. For example, the training data store may be configured to hold a certain amount of information. In some embodiments, the training data store may be configured to hold a certain number of examples. Each example may include a received prompt, the modified response, and the label. In some embodiments, prompt engine **106** may configure the training data store to hold data over a certain time window. For example, prompt engine **106** may configure the training data store to include examples from the last month, six months, or year. This may be beneficial because data in environment **100** may rapidly change, and it may be critical that LLM **212** trains on the latest information.

[0070] The modified response may be the response created by LLM **212**, modified by filter device **216**, and ultimately sent to client device **102**. As stated above, filter device **216** may have removed banned words and/or phrases as well as added required words and/or phrases to the response generated by LLM **212**.

[0071] At step **650**, prompt engine **106** trains the LLM using the training data store. In some embodiments, the LLM may be LLM **212**. Training LLM **212** may be useful to improve its performance with regard to generating accurate and precise responses to received prompts. Training may involve LLM **212** iterating over each stored prompt at the training data store, and generating a response.

[0072] LLM **212** may use the generated response to update its weights. The update may depend on the label corresponding to the example. For example, if a label indicates that the response was accurate with regard to the prompt, then ideally, the response generated by LLM **212** should be similar to or match the actual response sent to client device. Here, an error may be based on the difference between the generated response and the actual response. The error may be calculated by converting each response into a vector, and then calculating a similarity between the vectors. In this instance, the more similar the vectors, the smaller the error may be. The error may be used to update LLM **212** so that its generated response is more similar to the actual response. For example, backpropagation may be used to update LLM **212**.

[0073] Conversely, a label may indicate that the response sent to the client device was incorrect, or did not effectively respond to the prompt. In this example, LLM **212** should generate a response that is dissimilar to the actual response. Similar to the process above, an error may be calculated based on the similarity to the generated response and the actual response. In this case, the similarity should be low (e.g., the responses should be dissimilar) because the actual response did not address the prompt. Therefore, LLM **212** should be trained to respond differently to the prompt. The calculated error may be used to update LLM **212**.

[0074] It will be understood that the order of the above steps are merely exemplary, and the steps can be rearranged in any appropriate manner. Additionally, more or fewer steps may be included in the exemplary method consistent with the disclosure.

[0075] Various embodiments may be implemented, for example, using one or more well-known computer systems, such as computer system **700** shown in FIG. 7. One or more computer systems **700** may be used, for example, to implement any of the embodiments discussed herein, as well as combinations and sub-combinations thereof.

[0076] Computer system **700** may include one or more processors (also called central processing units, or CPUs), such as a processor **704**. Processor **704** may be connected to a communication infrastructure or bus **706**.

[0077] Computer system **700** may also include customer input/output device(s) **703**, such as monitors, keyboards, pointing devices, etc., which may communicate with communication infrastructure **706** through customer input/output interface(s) **702**.

[0078] One or more of processors **704** may be a graphics processing unit (GPU). In an embodiment, a GPU may be a processor that is a specialized electronic circuit designed to process mathematically intensive applications. The GPU may have a parallel structure that is efficient for parallel processing of large blocks of data, such as mathematically intensive data common to computer graphics applications, images, videos, etc.

[0079] Computer system **700** may also include a main or primary memory **708**, such as random-access memory (RAM). Main memory **708** may include one or more levels of cache. Main memory **708** may have stored therein control logic (i.e., computer software) and/or data.

[0080] Computer system **700** may also include one or more secondary storage devices or memory **710**. Secondary memory **710** may include, for example, a hard disk drive **712** and/or a removable storage device or drive **714**. Removable storage drive **714** may be a floppy disk drive, a magnetic tape drive, a compact disk drive, an optical storage device, tape backup device, and/or any other storage device/drive.

[0081] Removable storage drive **714** may interact with a removable storage unit **718**. Removable storage unit **718** may include a computer usable or readable storage device having stored thereon computer software (control logic) and/or data. Removable storage unit **718** may be a floppy disk, magnetic tape, compact disk, DVD, optical storage disk, and/or any other computer data storage device. Removable storage drive **714** may read from and/or write to removable storage unit **718**.

[0082] Secondary memory **710** may include other means, devices, components, instrumentalities or other approaches for allowing computer programs and/or other instructions and/or data to be accessed by computer system **700**. Such means, devices, components, instrumentalities or other approaches may include, for example, a removable storage unit **722** and an interface **720**. Examples of the removable storage unit **722** and the interface **720** may include a program cartridge and cartridge interface (such as that found in video game devices), a removable memory chip (such as an EPROM or PROM) and associated socket, a memory stick and USB port, a memory card and associated memory card slot, and/or any other removable storage unit and associated interface.

[0083] Computer system **700** may further include a communication or network interface **724**. Communication interface **724** may enable computer system **700** to communicate and interact with any combination of external devices, external networks, external entities, etc. (individually and collectively referenced by reference number **728**). For example, communication interface **724** may

allow computer system **700** to communicate with external or remote devices **728** over communications path **726**, which may be wired and/or wireless (or a combination thereof), and which may include any combination of LANs, WANs, the Internet, etc. Control logic and/or data may be transmitted to and from computer system **700** via communication path **726**.

[0084] Computer system **700** may also be any of a personal digital assistant (PDA), desktop workstation, laptop or notebook computer, netbook, tablet, smart phone, smart watch or other wearable, appliance, part of the Internet-of-Things, and/or embedded system, to name a few non-limiting examples, or any combination thereof.

[0085] Computer system **700** may be a client or server, accessing or hosting any applications and/or data through any delivery paradigm, including but not limited to remote or distributed cloud computing solutions; local or on-premises software (“on-premise” cloud-based solutions); “as a service” models (e.g., content as a service (CaaS), digital content as a service (DCaaS), software as a service (SaaS), managed software as a service (MSaaS), platform as a service (PaaS), desktop as a service (DaaS), framework as a service (FaaS), backend as a service (BaaS), mobile backend as a service (MBaaS), infrastructure as a service (IaaS), etc.); and/or a hybrid model including any combination of the foregoing examples or other services or delivery paradigms.

[0086] Any applicable data structures, file formats, and schemas in computer system **700** may be derived from standards including but not limited to JavaScript Object Notation (JSON), Extensible Markup Language (XML), Yet Another Markup Language (YAML), Extensible Hypertext Markup Language (XHTML), Wireless Markup Language (WML), MessagePack, XML User Interface Language (XUL), or any other functionally similar representations alone or in combination. Alternatively, proprietary data structures, formats or schemas may be used, either exclusively or in combination with known or open standards.

[0087] In some embodiments, a tangible, non-transitory apparatus or article of manufacture comprising a tangible, non-transitory computer useable or readable medium having control logic (software) stored thereon may also be referred to herein as a computer program product or program storage device. This includes, but is not limited to, computer system **700**, main memory **708**, secondary memory **710**, and removable storage units **718** and **722**, as well as tangible articles of manufacture embodying any combination of the foregoing. Such control logic, when executed by one or more data processing devices (such as computer system **700**), may cause such data processing devices to operate as described herein.

[0088] Based on the teachings included in this disclosure, it will be apparent to persons skilled in the relevant art(s) how to make and use embodiments of this disclosure using data processing devices, computer systems and/or computer architectures other than that shown in FIG. 7. In particular, embodiments can operate with software, hardware, and/or operating system implementations other than those described herein.

[0089] It is to be appreciated that the Detailed Description section, and not any other section, is intended to be used to interpret the claims. Other sections can set forth one or more but not all exemplary embodiments as contemplated by the inventor(s), and thus, are not intended to limit this disclosure or the appended claims in any way.

[0090] While this disclosure describes exemplary embodiments for exemplary fields and applications, it should be understood that the disclosure is not limited thereto. Other embodiments and modifications thereto are possible, and are within the scope and spirit of this disclosure. For example, and without limiting the generality of this paragraph, embodiments are not limited to the software, hardware, firmware, and/or entities illustrated in the figures and/or described herein. Further, embodiments (whether or not explicitly described herein) have significant utility to fields and applications beyond the examples described herein.

[0091] Embodiments have been described herein with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the

description. Alternate boundaries can be defined as long as the specified functions and relationships (or equivalents thereof) are appropriately performed. Also, alternative embodiments can perform functional blocks, steps, operations, methods, etc. using orderings different than those described herein.

[0092] References herein to “one embodiment,” “an embodiment,” “an example embodiment,” or similar phrases, indicate that the embodiment described can include a particular feature, structure, or characteristic, but every embodiment can not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it would be within the knowledge of persons skilled in the relevant art(s) to incorporate such feature, structure, or characteristic into other embodiments whether or not explicitly mentioned or described herein. Additionally, some embodiments can be described using the expression “coupled” and “connected” along with their derivatives. These terms are not necessarily intended as synonyms for each other. For example, some embodiments can be described using the terms “connected” and/or “coupled” to indicate that two or more elements are in direct physical or electrical contact with each other. The term “coupled,” however, can also mean that two or more elements are not in direct contact with each other, but yet still co-operate or interact with each other.

[0093] The breadth and scope of this disclosure should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. A system, comprising: a memory; and at least one processor coupled to the memory and configured to: identify a stored prompt based on a similarity value between a received prompt for a large language model (LLM) and the stored prompt, the stored prompt including a response generated by the large language model (LLM); and modify the response by applying a first rule associated with a first designated phrase, and a second rule associated with a second designated phrase, wherein the first designated phrase includes a banned phrase, and wherein the second designated phrase includes a selected phrase.
2. The system of claim 1, wherein the identifying further comprises: converting the received prompt to a vector representation; converting the stored prompt to a vector representation; and calculating the similarity value between the vector representation of the received prompt and the vector representation of the stored prompt.
3. The system of claim 2, wherein the similarity value is calculated via one of a cosine similarity, Euclidean similarity, or dot product similarity.
4. The system of claim 1, wherein identifying further comprises: determine a time period between the received prompt and the stored prompt is greater than a predefined time threshold; and in response to the determination, generating the response by applying the LLM to the received prompt.
5. The system of claim 1, wherein the banned phrase is removed from the response.
6. The system of claim 1, wherein the selected phrase is added to the response.
7. The system of claim 1, wherein the prompt is received from a client device, and the modification further comprises: identifying a previous prompt received from the client device; modifying the response based on the previous prompt from the client device.
8. A method, comprising: identifying a stored prompt based on a similarity value between a received prompt for a large language model (LLM) and the stored prompt, the stored prompt including a response generated by the large language model (LLM); and modifying the response by applying a first rule associated with a first designated phrase, and a second rule associated with a

second designated phrase, wherein the first designated phrase comprises a banned phrase, and wherein the second designated phrase includes a selected phrase.

9. The method of claim 8, further comprising: converting the received prompt to a vector representation; converting the stored prompt to a vector representation; and calculating the similarity value between the vector representation of the received prompt and the vector representation of the stored prompt.

10. The method of claim 9, wherein the similarity value is calculated via one of a cosine similarity, Euclidean similarity, or dot product similarity.

11. The method of claim 8, wherein the identifying further comprises: determining a time period between the received prompt and the stored prompt is greater than a predefined time threshold; and in response to the determination, generating the response by applying the LLM to the received prompt.

12. The method of claim 8, wherein the banned phrase is removed from the response.

13. The method of claim 8, wherein the selected phrase is added to the response.

14. The method of claim 8, wherein the prompt is received from a client device, and the modifying further comprises: identifying a previous prompt received from the client device; modifying the response based on the previous prompt from the client device.

15. A non-transitory computer-readable device having instructions stored thereon that, when executed by at least one computing device, cause the at least one computing device to perform operations comprising: identifying a stored prompt based on a similarity value between a received prompt for a large language model (LLM) and the stored prompt, the stored prompt including a response generated by the large language model (LLM); and modifying the response by applying a first rule associated with a first designated phrase, and a second rule associated with a second designated phrase, wherein the first designated phrase comprises a banned phrase, and wherein the second designated phrase includes a selected phrase.

16. The non-transitory computer-readable device of claim 15, wherein the operations further comprise: converting the received prompt to a vector representation; converting the stored prompt to a vector representation; and calculating the similarity value between the vector representation of the received prompt and the vector representation of the stored prompt.

17. The non-transitory computer-readable device of claim 16, wherein the similarity value is calculated via one of a cosine similarity, Euclidean similarity, or dot product similarity.

18. The non-transitory computer-readable device of claim 15, wherein the identifying further comprises: determining a time period between the received prompt and the stored prompt is greater than a predefined time threshold; and in response to the determination, generating the response by applying the LLM to the received prompt.

19. The non-transitory computer-readable device of claim 15, wherein the banned phrase is removed from the response.

20. The non-transitory computer-readable device of claim 15, wherein the selected phrase is added to the response.
